



LABORATORY MANUAL

SC2207/CZ2007: Introduction to Databases

Implementation of a Database Application

SCHOOL OF COMPUTER SCIENCE AND
ENGINEERING

NANYANG TECHNOLOGICAL UNIVERSITY

1. OBJECTIVES

Upon completion of the assignment, the student should be able to:

- a. Construct an entity-relationship model at a conceptual level.
- b. Map the model into a schema of a relational DBMS.
- c. Implement the given schema on a relational DBMS.
- d. Use a database language (SQL) to retrieval data from a relational DBMS.

2. INSTRUCTIONS

- a) **Team Formation**: This is a team-based assignment. Each team consists of **five to seven** members from your laboratory group, to be approved by your lab supervisor, to be formed during the Lab 1 session. The lab supervisor may add or remove members from your teams to ensure an even spread and mix of students in each team. The final members of your team must be submitted to the lab executive during the **Lab 1 session**.
- b) **Lab Submission**: There are five scheduled lab sessions for this team assignment. Laboratory sessions will start from **Week 3** for those scheduled on odd weeks, or **Week 4** for those scheduled on even weeks. For Lab 1, 3 or 5 submissions, do include a **cover page** indicating the team number and team members. Names of team members must appear as they do in student cards; do not shorten or use nicknames or aliases.
- c) **Lab Attendance**: Attendance is taken for the **first, third and fifth** lab sessions only. Attendance for the second and fourth lab sessions is not mandatory.
- d) **Fair Participation**: Each student is expected to make fair and equal contribution to EACH lab, and thoroughly understand the expectations of EACH lab. It should NOT be the case that a student contributes less to Lab X and make up for it by contributing more to Lab Y. Each submission needs to indicate contributions from each member. The final marks of a team member may be adjusted based on the team score and individual contribution. **Appendix C is to be submitted with each submission.**
- e) **Lab Supervisors**: For each lab session, there is a lab supervisor and a lab executive assisting you. The lab supervisor is a professor or a teaching assistant whom you may approach for clarifications on lab work, lab report submission, graded lab reports, etc. The lab executive is a technical staff whom you may approach for lab logistics (lab attendance, SQL Server account matter, lab submission deadline, computer problem, etc.).
- f) **AI tools**: NTU's policy on the use of AI tools: "The University requires students to (i) identify any generative AI tools used and (ii) declare how the tools are used in submitted work. Please note that

even with acknowledgement, copying of output generated by AI tools (in part or whole) **may still be regarded as plagiarism.** Appendix D is to be submitted with each submission.

3. INTRODUCTION

The assignment covers the portion of the course concerning data modelling, database design and implementation from the user's viewpoint. Thus, the assignment involves modelling as well as implementation aspects of the database course.

The overall aim of the laboratory is to develop an application based on a given data model using a given database management system. This exercise will bring you through a crucial first part of the life cycle of a database application. It is assumed that the data analysis has been performed. Note that this manual provides you with more information than is required for the first laboratory session, e.g., not all constraints can be modelled in the beginning but are included at a later implementation stage. In contrast you might require additional information for an understanding of the application. Proceed by stating your assumptions in written form and / or ask your laboratory supervisor.

4. DESCRIPTION OF THE ASSIGNMENT

The description of the application is given in Appendix A and B. This includes background and general requirements of the application, conceptual information about the system and its users as well as a list of SQL queries that must be fulfilled as a minimum. Note that teamwork is required. Each team will submit one solution. **No individual submission will be accepted.**

4.1 First Laboratory Session: Creating an ER Diagram

Appendix A gives conceptual information about the project obtained after a partial system analysis was performed. Based on the description, construct a suitable ER diagram. Analyze the choice of entity sets, different types of relationships required, the usage of weak entity sets, subclasses, etc. and compare them with alternative solutions from your team members. You need to submit the following, latest **three working days** after the first laboratory session, to the NTU Learn course site for your lab group:

- A PDF document of your ER diagram. A good ER diagram is one that is self-explanatory. If you believe certain parts of your ER diagram need explanation, you can include a written description (maximum one page). Combine both the ER diagram and the written explanation (if any) as a single PDF document, labeled as follows: Lab1_XXX_TeamY.pdf, where XXX is your lab group and Y is your team number. Marks are given for neat presentation of your ER diagram.

- Assessment for Lab 1 is based on whether the submitted ER diagram reflects correct understanding of ER diagram artefacts (entity sets, relationships, weak entities, subclasses, etc.) and whether they are used correctly and appropriately. Do note that not all information given in Appendix A can be represented in an ER diagram and that more than one ER diagrams are possible. It is part of the project work for your team to submit what your team deems to be the best ER diagram among all possible ones.

4.2 Second Laboratory Session: Finalization of the ER Diagram

There is no submission for the second laboratory session. In this lab, each team should finalize their database design based on the feedback received from their lab supervisor and prepare for Lab 3. Please note that the second laboratory session is a free access session, i.e., attendance is not mandatory.

4.3 Third Laboratory Session: Generation of Normalized Database Schema

In this lab, you convert the ER diagram into relational schema and ensure that the relations are at least in 3NF. Follow the general guidelines covered during the lectures and tutorials to produce suitably normalized relations. For each relation, the key(s), primary key, and functional dependencies must be specified. If a relation is generated due to the normalization of an original relation, then the normalization steps must be presented. You need to submit the following, latest **three working days** after the third laboratory session, to the NTU Learn course site for your lab group:

- A PDF document of the normalized database schema and FDs associated with each relation. Label the PDF document as: Lab3_XXX_TeamY.pdf, where XXX is your lab group and Y is your team number. If a relation that is created from the ER diagram violates 3NF, then this should be highlighted along with the decomposed normalized relations. Note that for this lab, no SQL code should be submitted. Hence, the structure of your solution shall be similar to the following example:

R1(A, B, C, D)

Keys: AB, AD

Primary Key: AB

FDs: $AB \rightarrow CD$, $A \rightarrow D$

The relation is in 3NF. (If relation is not in 3NF, perform the steps of the 3NF normalization.)

- Assessment for Lab 3 is based on whether the submitted report reflects correct understanding of keys in relations, identification of appropriate functional dependencies in each relation, how normalized relations are formed, and whether the normalizations are correctly and

appropriately performed. Do note that in your final set of relations, the keys and functional dependencies in each relation may not be explicitly given in the description in Appendix A.

4.4 **Fourth Laboratory Session: Implementation of the database schema**

There is no submission for the fourth laboratory session. In this lab, the finalized database schema is to be implemented using SQL DDL commands. Your tables should be appropriately populated with sufficiently realistic records using SQL INSERT statements so that your query solution for Appendix B results in some meaningful output records (3 to 5) for each query. Your implementation should clearly incorporate the primary and foreign keys, data types, and any form of constraints. The lab provides MS SQL Server software for your implementation. You should start to work on the queries in Appendix B.

Please note that the fourth laboratory session is a free access session, i.e., attendance is not mandatory.

4.5 **Fifth Laboratory Session: Final demonstration**

In this lab, the implementation obtained from the previous laboratory session must now be extended to provide SQL query solutions for the queries in Appendix B. **At the end of the lab session**, you need to submit a single PDF document to the NTU Learn course site for your lab group containing the followings:

- SQL DDL commands for table creation (from Lab 4).
- SQL statements to solve the queries in Appendix B and additional queries (if any). Each query should be immediately followed by the query output. Briefly explain how the output is obtained.
- A printout of all table records.
- Description of any additional effort made.

Label the PDF document as: Lab5_XXX_TeamY.pdf, where XXX is your lab group and Y is your team number. You should prepare the PDF document in advance before coming to the lab. Some DDL commands may look like this:

```
CREATE TABLE name (  
    attr1 datatype NOT NULL,  
    attr2 datatype,  
    ...  
    PRIMARY KEY (attr1),  
    FOREIGN KEY (attr3) REFERENCES name(attr1)  
    ON DELETE ... ON UPDATE ...,
```

);

In addition to the PDF document, you are to capture **screen recording** of query execution as a mp4 video file. For each query in Appendix B and additional queries, first show the SQL statement, then execute the query and show the query results, all recorded as a mp4 video file. Each query video should be no more than 30 seconds and labeled as: Lab5_XXX_TeamY_Q#.mp4 where # is the query number. Zip the PDF and all mp4 files into one single ZIP file.

During the lab session, you may be given additional queries to solve. In addition, your lab supervisor may require in-person live demonstration and Q&A. All team members are to actively contribute during the demonstration session and be familiar with **all aspects** of the project. No slide presentation is required.

APPENDIX A: APPLICATION DESCRIPTION

The tourism agency of ABC country is developing an app to attract and assist people visiting malls, restaurants, parks, tourist attractions, etc. After a user signed up for an account with the app, if the user visits the malls and restaurants recommended by the app within a certain time period, the user gets a voucher. This can be a cash voucher or a purchase voucher. This allows the app not only to gain traction among people but also gather information about people's visiting behaviour, which the tourism agency can use to roll out more schemes to boost the tourism and retail economy. Here is more information about the app and its usage scenarios.

1. When a user signs up, the app requests for the user's phone number, date of birth (optional), name (optional), and gender (optional). The phone number is used primarily as the user's login identifier. Subsequent user information such as email address, home address, monthly income, etc. may be requested and recorded in the app system.
2. Malls generally have diverse retail shops to satisfy people's browsing and impulsive buying crave, apart from acquiring essential necessities. Some malls are from a mall chain. In Singapore, CapitaLand Mall is a mall chain that owns and manages many malls such as Bedok Mall, Tampines Mall, IMM Building, Junction 8, etc. The app system records information about mall chain, individual malls from mall chains, and individual malls that do not belong to any mall chain. The app system also records information about shops in a mall.
3. Malls also have restaurants and cafeterias where people rest and dine. Some restaurants are from a restaurant chain. For example, MacDonalds and Burger King are restaurant chains with outlets throughout multiple malls. The app system records information about restaurant chain, restaurant outlets from restaurant chains, and individual restaurants that do not belong to any restaurant chain.
4. The app algorithmically recommends malls and restaurants to users and send them notifications of these recommendations. Once notifications are received, a user may then visit the recommended malls and restaurants within the designated time period and stands to receive vouchers from the app. If a user visits a recommended mall, he/she may receive a purchase voucher where it can be redeemed at shops in the malls. If a user visits a recommended restaurant, he/she may receive a cash voucher where it can be used to offset the bill at the restaurant. Alternatively, the user may visit malls and restaurants without using the app and receive no vouchers at all. The app system records information about vouchers in general, and vouchers issued and used. The system tracks the usage and status of each voucher. Its status could be "allocated", "redeemed", or "expired".
5. The app has an optional ChatGPT-like mode where users may interact with the app more colloquially, in English, Chinese, Malay, Tamil, and Singlish. Based on users' input prompts, the app learns and personalizes future conversations according to users' preferences. All learned information is currently stored with ChatGPT and the tourism agency is trying to have the learned information stored locally within its database system in future.

6. The app proposes day packages for users. A day package is an itinerary of malls that will be visited within a day. Users are shuttled by a bus from one mall to another throughout the day's itinerary. Users who signed up for a day package are given discount vouchers for dining in those malls in the itinerary. Day packages are particularly popular among groups of friends and/or family members, who take these packages as one-day tours, with discounted meals.
7. Users are allowed to file complaints on any shops and restaurants. For instance, the user may complain that the restaurant is dirty, or its attendant's service is poor. For shops, the user may complain that the prices of items are too high, or the shop provided bad service. After a complaint is filed, the customer can check on the app the status of his/her complaint. The status could be "pending", "being handled", or "addressed".
8. From the information collected of app users, the system applies artificial intelligence/machine learning (AI/ML) techniques to categorize and predict user behaviour in shopping activity and dining activity. For instance, AI/ML techniques may be used to provide insights into correlation between user's age and the types of shops he/she shops and the amount of money spent; between user's gender/age and patterns of shopping in malls.
9. Sometimes, users visit malls with their family members or friends or colleagues. The app allows a user to indicate which other users are related to them when visiting. For instance, User A indicates during his visit or dining in the app that User B is his father; User C is his classmate; User D is his colleague; User E is a follower of User A's Instagram account, etc. These other users must also be signed in to the app as well. Using this information, the app system may recommend group discount vouchers for active users.
10. The database should support the queries listed in Appendix B.

Note that the information above may not be complete. Some aspects of the database application's details may have been omitted. It is expected that you come up with their own solution(s) in case of inconsistencies or missing information. However, you have to keep track of these aspects and explain your assumptions in your submitted report.

A

APPENDIX B: QUERIES

1. Find the most popular day packages where all participants are related to one another as either family members or members of the same club.
2. Find families who frequently shopped and dined together, with or without day packages. As part of your output, indicate whether these families use day packages or not. “frequently” means at least 50% of the time.
3. What are the most popular recommendations from the app regarding malls?
4. Compulsive shoppers are those who have visited a certain mall more than 5 times within a certain period of time. Find the youngest compulsive shoppers and the amount they spent in total during December 2023.
5. Find users who have dined in all the restaurants in some malls, but have never dined in any restaurants in some other malls.
6. What are the top 3 highest earning malls and restaurants?

Note: For all the above queries, you must populate your tables with sufficient data to generate concrete result outputs. No query should generate a NIL (empty) output.

APPENDIX C: INDIVIDUAL CONTRIBUTION FORM

Full Name	Individual Contribution to Lab 1 Submission	Percentage of Contribution	Signature

Full Name	Individual Contribution to Lab 3 Submission	Percentage of Contribution	Signature

Full Name	Individual Contribution to Lab 5 Submission	Percentage of Contribution	Signature

APPENDIX D: USE OF AI TOOL(S) IN LAB WORK

Each team member should indicate either A or B:

A. I affirm that my contribution(s) to the lab work is my own, produced without help from any AI tool(s).

B. I affirm that my contribution(s) to the lab work has been produced with the use of AI tool(s).

Team member (full name)	Signature	Date	A or B

By signing this form, you declare that the above affirmation made is true and that you have read and understood NTU's policy on the use of AI tools.

If any team member answered B, the team member(s) must indicate and replicate the table below for every instance AI tool(s) is used:

Name of AI tool	< For example, ChatGPT >
Input prompt	< Insert the question that you asked ChatGPT >
Date generated	
Output generated	< Insert the response verbatim from ChatGPT >
Output screenshots	
Impact on submission	< Briefly explain which part of your submitted work was ChatGPT's response applied >